2024

# Processing Checks Up to 80% Faster Using AWS Step Functions Distributed Map with Capital One

Learn how Capital One in financial services reduced check processing time and increased productivity using AWS Step Functions Distributed Map.

**Overview** | **Opportunity** | **Solution** | **Ou**



**Upto 80%**

**25x**

Reduction in overall processing time

Number of concurrent workflows launched

**1000s**

Workflows closed in parallel

## Overview

Financial services company Capital One is always looking for ways to streamline processes and use the latest technology from Amazon Web Services (AWS). When AWS released a new feature for AWS Step Functions, a visual workflow service for distributed applications, Capital One wanted to use it right away to expedite its check clearing application. By optimizing the use of AWS Step Functions, Capital One reduced the processing time for its check clearing application by up to 80 percent and greatly improved analyst productivity.


## Opportunity | Using AWS Step Functions Distributed Map to Increase Productivity for Capital One

Founded in 1994, Capital One provides credit cards, checking accounts, savings accounts, auto loans, and more. The company is committed to financial inclusion and serves more than 100 million customers across a diverse set of businesses.

Capital One hosts its check clearing application on AWS and uses AWS Step Functions to orch loads are processed by AWS Lambda, a serverless compute service. The company processes thousan y day and uses the digital file storage capacity offered by Amazon Simple Storage Service (Amazon S3), an object storage service built to retrieve virtually any amount of data from anywhere.

To orchestrate the check clearing application, Capital One was successfully using AWS Step Functions in the inline map state, which runs the same processing steps for multiple entries in a dataset with up to 40 parallel iterations. While most checks clear automatically, analysts need to manually review a small but consistent portion of checks before the close of each business day. Soon after AWS released the option to use AWS Step Functions Distributed Map, which can launch up to 10,000 parallel workflows to process data, Capital One seized the opportunity to use parallel processing to further streamline its workflows and identify checks needing manual review faster. After 10 weeks of planning and testing a successful proof of concept, the company switched to using AWS Step Functions Distributed Map in production with no major architecture changes or downtime.

In addition to adopting AWS Step Functions Distributed Map soon after its release, Capital One has worked closely alongside teams at AWS to preview, test, and provide feedback for other upcoming features. “We’re constantly working alongside the AWS team to see what’s coming on the road map and how we can use it,” says Sushma Onkar, distinguished engineer at Capital One. “That helps us innovate faster.”

## Solution | Reducing Processing Time by Up to 80 Percent and Freeing Up Time for Analysts Using AWS Step Functions

Using AWS Step Functions Distributed Map, Capital One reduced the overall processing time for launching and closing workflows by 75–80 percent. The company can now launch 25 times the number of concurrent workflows for processing instead of working in batches of 40. Using AWS Step Functions Distributed Map, Capital One can also close thousands of workloads in parallel, which significantly reduces the overall check processing time. “The time to launch and close processes was drastically improved,” says Onkar. “Analysts see updates to their queues faster than before.” Even though check volumes fluctuate throughout the week, the application can scale to meet the demands of the parallel workflows using AWS Lambda. “When we implemented AWS Step Functions Distributed Map, we could benefit from the serverless scaling capabilities of AWS Lambda and process a large volume of checks faster,” says Onkar.

Because workflows are happening in parallel, the application completes processing faster, and analysts can get started on manual review sooner. These productivity improvements free up time for analysts [illegible] r tasks. “The teams have been very happy with the change to use AWS Step Functions Distribut[illegible] kar.

Previously, Capital One occasionally experienced throttling exceptions if the check clearing application experienced a high volume of API calls simultaneously. Using the automated retry feature of AWS Step Functions, Capital One avoids these exceptions and further improves productivity.

## Outcome | Exploring Additional Innovative Use Cases Using AWS Services

Capital One continues to work closely alongside teams at AWS to explore new services and features that can improve workflows. Other teams within the company are also considering using AWS Step Functions Distributed Map to expedite data processing workflows. "Using serverless functionality from AWS, we can explore and adapt to constant innovation, which improves our productivity and the customer experience," says Onkar.

## About Capital One

Founded in 1994, financial services company Capital One provides credit cards, checking accounts, savings accounts, auto loans, and more to more than 100 million customers.

## AWS Services Used

### AWS Step Functions

AWS Step Functions is a visual workflow service that helps developers use  AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

Learn more »

### AWS Lambda

AWS Lambda is a compute service that runs your code in response to  events and automatically manages the compute resources, making it the  fastest way to turn an idea into a modern, production, serverless applications.

Learn more »

### Amazon Simple Storage Service

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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